

# WARNLY

Next-Generation Hyper-Local Convective Hazard Early Warning & Precision Lightning Defense System: Feasibility, Critical Bottleneck Resolutions, Social Impact, and Production Architecture

## DOCUMENT CLASSIFICATION

Technical & Product White Paper

## DOCUMENT RELEASE

Version 3.2 (Production Hardened)

## FOCUS DOMAIN

Severe Convective Storms & Lightning

### 🎯 Core Mission

Transforming unpredictable atmospheric convective threats into deterministic, actionable personal survival windows. Warnly delivers 10–25 minutes of advance lead-time for approaching storm cells before lightning breaches the 10 km safety perimeter.

### 🔬 Physics-True Calibration

Resolving warning fatigue through a disciplined two-tier model: separating general atmospheric convective build-up (Watch) from real-time spatial storm cell advection tracking (Tactical Evacuation).

### ⚡ Convective Focus

Laser-focused on deep moist convection hazards: precision lightning strike tracking, microbursts, and convective flash flood downpours—eliminating architectural bloat from disjointed seismic feeds.

### 🛡️ Hardened Notification Architecture

A commercially viable alerting pipeline utilizing iOS Time-Sensitive notifications, persistent Live Activities, and Android high-priority channels to guarantee urgent delivery without entitlement roadblocks.

## 1 Executive Summary & Physical Foundations

### ATMOSPHERIC CONVECTIVE DYNAMICS

**Warnly** is an advanced, hyper-local life safety platform engineered to protect outdoor workers, athletes, and families from severe convective storms and lightning strikes. While national weather bureaus (e.g. IMD, NOAA) provide broad regional forecasts across thousands of square kilometers ("thunderstorms likely over district in next 3 hours"), lightning strikes are localized, violent, and rapid. Warnly transforms raw atmospheric telemetry into tactical, perimeter-based survival windows.

**The Fundamental Atmospheric Reality:** Broad regional weather alerts create severe warning fatigue. People sitting under sunny skies ignore alerts because the storm is 45 km away in the same district. Warnly solves this by establishing individual geodesic radii centered on the user's live coordinates.

### 1.1 Storm Advection vs. Convective Initiation (The Lead-Time Truth)

To maintain absolute scientific integrity, Warnly distinguishes between the two distinct physical ways severe thunderstorms impact an individual:

#### A. Advecting Storm Cells (~85% of Cases)

10-25 MIN  
ADVANCE

A convective storm is already active 20–35 km away and traveling toward the user at 30–45 km/h. By tracking real-time strike vectors and radar reflectivity cores, Warnly calculates the precise time-to-breach, delivering a **genuine 10–25 minute evacuation window** before the first strike enters the 10 km danger ring.

#### B. In-Situ Convective Initiation (~15% of Cases)

WATCH  
LEVEL

A cloud rapidly develops directly overhead on a hot afternoon. No algorithm on Earth can predict the exact second or ground point of the very first discharge before it occurs. For this scenario, Warnly delivers an **Atmospheric Convective Watch** based on high CAPE (> 1500 J/kg) and negative Lifted Index, prompting users to wrap up outdoor work without false alarms.

### 1.2 Core Capabilities Overview

- Precision Geodesic Safety Rings:** Calculates Great-Circle distance to strikes in sub-milliseconds. 10 km Critical Danger Ring (derived from thunder's 30-second acoustic travel boundary) and 15 km Advisory Ring.
- The Automated 30-30 Sheltering Timer:** Over 30% of lightning casualties occur when people leave shelter prematurely. Warnly runs a 30-minute countdown that automatically resets to 30:00 with each new strike inside 10 km.
- Focused Convective Hazard Scope:** Concentrated entirely on Deep Moist Convection: cloud-to-ground lightning, microburst wind gusts, and sudden convective flash downpours.



## 2 Comprehensive Feasibility & Data Architecture

TECHNICAL, COMMERCIAL DATA & ECONOMICS

### 2.1 Compliant Data Ingestion Architecture

Relying on hobbyist open networks (e.g. Blitzortung) for commercial products violates their Terms of Service and creates severe legal liability. Warnly resolves this through a legitimate, tiered data ingestion pipeline:

| Data Source Tier                                         | Providers & Technology                                                                                                                                                                                                                      | Licensing & Commercial Status                                                                                                                               |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tier 1: Free Public Domain (Zero Licensing Cost)</b>  | <b>NOAA GOES-16 &amp; GOES-18 GLM:</b><br>Geostationary Lightning Mapper optical feeds. Detects in-cloud and cloud-to-ground pulses every 20 seconds.<br><b>Open-Meteo &amp; ECMWF:</b> Convective indices (CAPE, Lifted Index, rain rate). | <b>100% Legal &amp; Free for Commercial Use.</b> US Government public domain datasets hosted on AWS Open Data / Google Cloud BigQuery with zero usage fees. |
| <b>Tier 2: Commercial Telemetry (Funded by B2B SaaS)</b> | <b>Vaisala GLD360 / Earth Networks (ENTLN):</b><br>Global precision VLF/LF Time-of-Arrival sensor networks delivering sub-second ground stroke telemetry with 95%+ detection efficiency.                                                    | <b>Commercial API License.</b> Funded via enterprise SaaS subscriptions (golf resorts, construction companies, airports paying \$150–\$500/month).          |

### 2.2 Technical Feasibility & Sub-Second Processing at Low Cost

The client assumption that sub-second real-time tracking requires millions in supercomputing hardware is an architectural misconception:

- **Client-Side Edge Computing:** The central server does *not* compute distances for 1,000,000 individual users. The server simply broadcasts: Strike at (lat, lon) to subscribed spatial rooms. The user's smartphone processor computes the Haversine distance in **under 20 microseconds (0.00002 seconds)** with negligible battery draw.
- **Spatial Partitioning (Uber H3 Indexing):** The globe is divided into discrete hexagonal cells. Strike coordinates are hashed to an H3 cell in 0.001 ms, pushing events only to connected clients in the corresponding spatial bucket.
- **Enterprise Cloud Free Tiers:** A cluster running on Oracle Cloud Always-Free Tier (4 ARM Ampere Cores, 24 GB RAM) can easily hold 50 million strike records in an in-memory Redis cache with zero server cost.

### 2.3 Financial Viability: Freemium Safety Model

**Sustainable B2B Unit Economics:** The free tier (universal human safety) provides core lightning alerts using public domain satellite data (NOAA GOES GLM) and open meteorological feeds. Commercial enterprise subscriptions (construction site crane lockouts, athletic leagues, golf courses) pay for high-precision commercial VLF ground-stroke network feeds, creating a profitable, legally compliant enterprise.

3

Critical Bottlenecks & Strategic Resolutions

HARDENED ENGINEERING TRADE-OFFS

A rigorous, battle-tested system must confront real-world operating platform restrictions. Here is how Warnly resolves the four primary engineering bottlenecks:

| Identified Bottleneck                            | The Real-World Industry Constraint                                                                                                                              | Warnly's Hardened Strategic Solution                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apple Critical Alerts Restriction             | Apple rejects ~99% of commercial apps requesting the Critical Alerts entitlement (reserved strictly for medical systems and certified first responders).        | <b>Dual-Track Approach:</b> <ul style="list-style-type: none"> <li><i>Consumer Track:</i> Utilizes iOS 15+ <b>Time-Sensitive Notifications</b> (breaks through Focus/Sleep modes without special entitlement) + <b>Live Activities</b> on lock screen.</li> <li><i>Enterprise/B2G Track:</i> Partner with accredited municipal emergency management bodies to co-sign the Apple Critical Alerts application.</li> </ul> |
| 2. Lead-Time vs. Zero False Alarms Contradiction | Atmospheric physics cannot provide both 20-minute advance lead-time AND 0% false alarms for the first strike of a newly initiating cell overhead.               | <b>Two-Tier State Machine:</b> <ul style="list-style-type: none"> <li><i>Convective Watch:</i> Triggered by high CAPE/negative LI (advisory only, no panic).</li> <li><i>Tactical Evacuation:</i> Triggered by real upstream strikes approaching at velocity <math>V_{\text{vector}}</math> or radar core exceeding 45 dBZ. 100% honest physics.</li> </ul>                                                             |
| 3. Open Data Commercial Licensing Bans           | Blitzortung terms strictly ban commercial SaaS usage. Building a business on unauthorized open feeds invites cease-and-desist orders.                           | <b>Legitimate Hybrid Feed:</b> Free consumer tier runs on public domain NOAA GOES-16/18 GLM satellite telemetry; commercial B2B subscribers directly fund licensed commercial feeds (Vaisala GLD360 / Earth Networks).                                                                                                                                                                                                  |
| 4. Disparate Modality Bloat (Seismic / GLOF)     | Bundling USGS earthquake feeds (which update with 1-5 min latency) and GLOF into a lightning app causes architectural bloat and destroys technical credibility. | <b>Focused Scope:</b> Decoupled entirely from seismic alerts. Warnly is 100% dedicated to <b>Deep Moist Convection Hazards</b> (Lightning, Microbursts, and Convective Flash Downpours), where data models share identical physics.                                                                                                                                                                                     |

### 3.1 Realistic Notification Matrix Across Mobile OS Platforms

iOS Architecture (Apple Ecosystem)

APPROVED PATH

- `UNNotificationInterruptionLevelTimeSensitive` : Guaranteed high-priority delivery through Focus Modes and Scheduled Summaries.
- `ActivityKit (Live Activities)` : Displays live radar distance and 30-30 countdown timer directly on the Lock Screen and Dynamic Island.

Android Architecture (Google Ecosystem)

NATIVE OVERRIDE

- `NotificationManager.IMPORTANCE_HIGH` : Heads-up pop-up display.
- `USE_FULL_SCREEN_INTENT` : Wakes phone screen and mounts emergency intrusion modal even when phone is locked or on DND.

## 4 Societal Impact & Quality of Life

TRANSFORMING SAFETY CULTURE & COMMUNITY RESILIENCE

### 4.1 The Silent Epidemic: Lightning Casualties & Long-Term Trauma

Lightning is one of the deadliest yet most preventable meteorological hazards on Earth. According to the World Meteorological Organization (WMO):



### 4.2 High-Risk Demographics & Real-World Use Cases

| User Segment                        | Exposure Profile                                                  | Warnly Societal & Life-Saving Impact                                                                                                                                                  |
|-------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agriculture & Field Laborers        | Extreme (Open fields, metal tools, tractors, isolated trees)      | Farmers account for >50% of lightning deaths globally. Warnly's 15-minute advance advection warning provides the necessary window to dismount tractors and reach hardened shelter.    |
| Construction & Heavy Industry       | High (Tower cranes, metal scaffolding, elevated ironwork)         | OSHA mandates suspending crane and elevated operations when lightning enters 10 miles. Warnly provides auditable compliance alerts, protecting workers from electrocution.            |
| Schools, Athletics & Outdoor Sports | High (Open fields, stadiums, swimming pools, playgrounds)         | Enforces athletic lightning safety rules. The automated 30-30 timer prevents coaches from prematurely sending children back onto sports fields while secondary strikes remain active. |
| Mariners, Boaters & Water Sports    | Extreme (Open water conductivity, masts acting as lightning rods) | Water conducts current across vast distances. Warnly warns boaters 20 minutes before storm arrival, allowing time to dock or seek enclosed cabin shelter.                             |

### 4.3 The Psychological Shift: From Anxiety to Deterministic Protocol

Thunderstorm anxiety stems from unpredictability. When skies darken, individuals guess whether danger is 2 km or 20 km away. Warnly replaces ambiguity with transparent telemetry: exact distance to nearest strike, storm velocity vector, and an unambiguous countdown timer, fostering a disciplined culture of personal safety.

### 4.4 Economic Return on Investment (SROI)

**Preserving Human Capital & Economic Assets:** Beyond preventing fatalities, avoiding lightning trauma eliminates massive lifelong healthcare costs (cardiac rehabilitation, neurological treatment, chronic pain management). For commercial enterprises, early alerts prevent multi-million-dollar wrongful death lawsuits and catastrophic machinery fires.

## 5 System & Functional Requirements Specification

### CORE FUNCTIONAL & NON-FUNCTIONAL SPECIFICATIONS

#### 5.1 Core Functional Requirements (FR)

| Identifier                 | System Capability               | Engineering Specification & Acceptance Criteria                                                                                                                                            |
|----------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01: Spatial Rings       | Geodesic Safety Radii           | Compute Great-Circle distance to nearest strike in real time. Flag Danger when distance $\leq 10.0$ km; flag Advisory when $10.0 \text{ km} < \text{distance} \leq 15.0$ km.               |
| FR-02: Two-Tier Risk       | Advection vs Watch Gating       | Display "Convective Watch" if CAPE $> 1500 \text{ J/kg}$ without active strikes. Escalate to "Tactical Evacuation" when upstream strikes are detected within 35 km moving toward the user. |
| FR-03: Shelter Timer       | Automated 30-30 Countdown       | Display a 30:00 countdown clock upon entering Danger state. Automatically reset timer back to 30:00 upon each subsequent strike detected within 10 km.                                     |
| FR-04: Emergency Intrusion | Full-Screen Danger Overlay      | Mount an un-dismissible high-contrast danger card across the viewport when strikes breach 10 km, displaying immediate indoor shelter steps.                                                |
| FR-05: Acoustic Beacon     | Software Siren Synthesizer      | Synthesize an 880Hz / 440Hz bi-tonal emergency siren via Web Audio oscillators with zero network download requirements; include a Morse SOS optical strobe.                                |
| FR-06: Shelter Finder      | High-Ground & Hardened Shelters | Query nearest verified civil shelters and hardened buildings with distance, compass bearing, and walking directions cached locally on device.                                              |
| FR-07: Family Shield       | Multi-Zone Perimeter Rings      | Allow users to configure and monitor up to 10 distinct geographic zones simultaneously, issuing discrete alerts if lightning breaches any monitored zone.                                  |

#### 5.2 Non-Functional & Performance Requirements (NFR)

##### Latency & Ingestion Speed

&lt; 1.5 SECONDS

Total latency from physical lightning ground-contact to mobile notification delivery must not exceed 1.5 seconds under standard cellular conditions.

##### Availability & Reliability

99.99% UPTIME

API gateways and WebSocket pub/sub brokers must maintain 99.99% uptime during regional severe convective weather outbreaks.

##### Battery Consumption

&lt; 2% PER 24 HOURS

Adaptive GPS power governor must maintain background battery consumption under 2% per 24 hours during normal Safe states.

##### Privacy & Security

ZERO TRACKING

User GPS coordinates must be processed ephemerally in RAM for spatial distance calculations and never persisted in permanent database logs.

#### 5.3 Offline Reliability & Failover Standards

When extreme convective storms damage cellular base stations, the mobile application must degrade gracefully: safety protocols, shelter directions, the 30-30 timer, and synthesized acoustic sirens must remain 100% operational offline.

6

Proposed Technology Stack & Architecture

RECOMMENDED SYSTEMS ENGINEERING BLUEPRINT

6.1 Production Architecture Topology

Warnly is designed as a distributed, high-throughput stream processing pipeline separating telemetry ingestion, spatial discrete indexing, and edge client evaluation:

| System Tier              | Recommended Technologies                           | Engineering Justification & Capabilities                                                                                                                      |
|--------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobile & Web Client      | React Native / Flutter + Progressive Web App (PWA) | Cross-platform deployment. React Native / Flutter provides native background notification channels and Live Activities; PWA enables zero-friction web access. |
| Geospatial UI & Maps     | Mapbox GL / Leaflet with SVG Canvas Overlay        | Hardware-accelerated map rendering capable of visualizing concentric 10km/15km safety circles, heatmaps, and age-coded strike vectors without UI lag.         |
| Real-Time Gateway        | Go (Gin/Fiber) or Node.js (Fastify + Socket.io)    | Handles 100,000+ concurrent bi-directional WebSocket connections per node. Dispatches low-bandwidth telemetry payloads to connected spatial rooms.            |
| In-Memory Cache & Broker | Redis (Pub/Sub + Redis Streams)                    | Sub-millisecond broker distributing live lightning events from ingestion workers to gateway nodes; caches regional weather grid calculations.                 |
| Spatial Compute Engine   | Python (FastAPI, Geopy, Shapely, NumPy) or Go      | Calculates spatial geometry intersections, storm direction vectors, convective grid interpolation, and safe evacuation route calculations.                    |
| Database & Storage       | PostgreSQL 16 + PostGIS extension                  | Stores verified safe shelter coordinates, emergency contact lists, historical storm logs, and multi-zone Family Shield polygons with spatial indexing.        |
| Acoustic & Hardware      | Web Audio API OscillatorNode / Native Audio DAC    | Synthesizes bi-tonal alarm sirens (880Hz / 440Hz) entirely in software memory, eliminating dependencies on external audio asset downloads.                    |

6.2 Multi-Tier Telemetry Ingestion Pipeline

Tier 1: Free Public Ingestion

OPEN ACCESS

- **NOAA GOES-16/18 GLM:** Continuous geostationary satellite optical pulse detection. 100% free for commercial use.
- **Open-Meteo NWP:** ECMWF/GFS high-resolution atmospheric instability indices (CAPE, Lifted Index).

Tier 2: Commercial Telemetry

ENTERPRISE FUNDED

- **Vaisala GLD360 / Earth Networks:** Sub-second ground stroke VLF/LF Time-of-Arrival telemetry with 95%+ detection efficiency.
- Subsidized directly via B2B enterprise subscription revenue.

6.3 Stream Processing & Edge Dispatch Flowchart



## 7 Strategic Execution Roadmap & Team Framework

### PHASING, OPERATIONAL STANDARDS & CONCLUSION

#### 7.1 Four-Phase Implementation Roadmap

| Phase   | Milestone Target          | Key Deliverables & Architectural Focus                                                                                                                                                    | Target Window |
|---------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Phase 1 | Core Engine & Open Ingest | Integration with public domain NOAA GOES GLM telemetry, Haversine geodesic safety perimeters (10km/15km), Open-Meteo convective weather suite, 30-30 timer, and Web Audio siren.          | COMPLETED     |
| Phase 2 | Native Mobile & B2B Feeds | React Native / Flutter apps with iOS Time-Sensitive Notifications, Live Activities Lock Screen widget, Android High-Priority channels, and licensed commercial lightning API integration. | Q3 - Q4 2026  |
| Phase 3 | Enterprise Compliance Hub | Operations console for construction firms, golf courses, airports, and athletic leagues. Automated IoT hardware relays triggering field horn sirens and crane lockouts.                   | Q1 - Q2 2027  |
| Phase 4 | Civil Defense & LoRa Mesh | Co-branded deployments with accredited public safety bodies (unlocking full Apple Critical Alerts entitlement) and LoRa mesh radio broadcasting for zero-cellular rural regions.          | Q3 - Q4 2027  |

#### 7.2 Core Operational Principles for the Engineering Team

- **The Scientific Integrity Mandate:** Never claim to predict the millisecond of an initiating strike before it happens. Maintain the strict distinction between convective instability watches and deterministic advective breach alarms.
- **Sub-Second Latency Obsession:** Treat every millisecond of alert delivery latency as life-critical. Offload distance calculations to client-side edge computing to guarantee sub-second user responses without server bottlenecking.
- **Offline-First Resilience:** Every safety protocol, emergency shelter directory, countdown timer, and acoustic siren beacon must function with zero network access during severe storm cell collapse.
- **Strict Privacy Guarantees:** Process user location coordinates ephemerally in RAM. Never persist or monetize real-time user movement profiles.

#### 7.3 Executive Conclusion & Mission Statement

**The Warnly Mandate:** Lightning strikes in microseconds; convective storms evolve in minutes. Warnly proves that disciplined software engineering, client-side edge computing, and space-based telemetry can provide citizens with the deterministic survival window they need to reach shelter safely. Every line of code written for this platform directly safeguards human life.